



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251

Spring Semester 2025

Instructor: Usman Salahuddin

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Quiz 01 [Due: Feb 13th]

Presentations (12.5%) (Feb 18th)

- Make a group of 3
- Pick a country (except Pakistan)
- Study their water and sewerage management system
- Make a 7 minutes presentation (3.5 minutes for each student)
- 2-3 minutes Q&A

Grading criteria:

[Time (2.5%), Quality of presented material (5%), Presentation (2.5%), Q&A response (2.5%)]

Research Project (30%)

- Group activity (Groups of 3)
- Students review literature and conduct field work to understand local water related problems and propose possible solutions.

	Grade % (Total 30%)
Proposal	2.5
Introduction	2.5 [Due: Feb 13]
Literature review	2.5 [Due: Feb 25]
Primary research	7.5
Presentation	5
Final Report	10

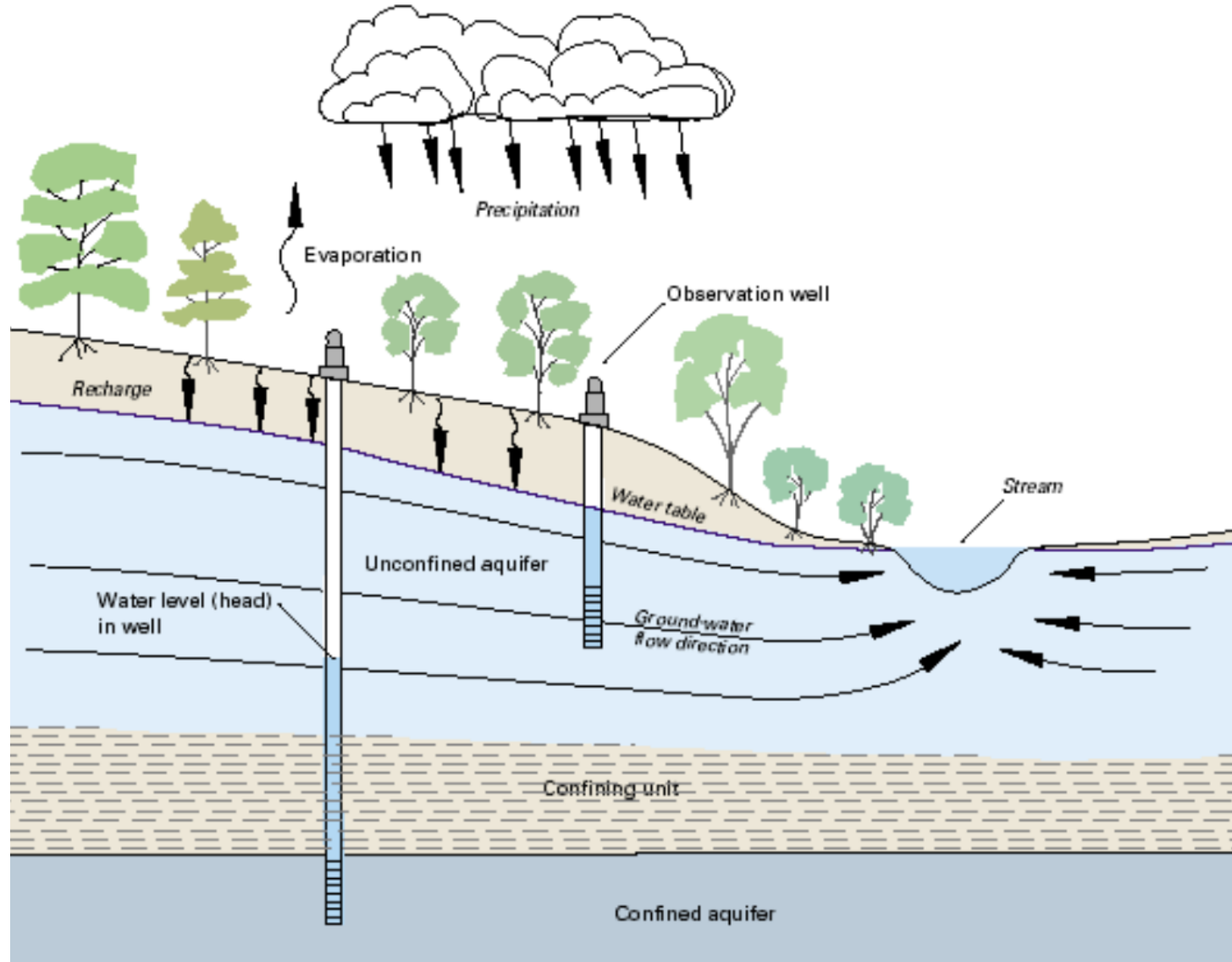
Introduction (2.5%) [Due: Feb 13]

Write a two-page introduction for your project.

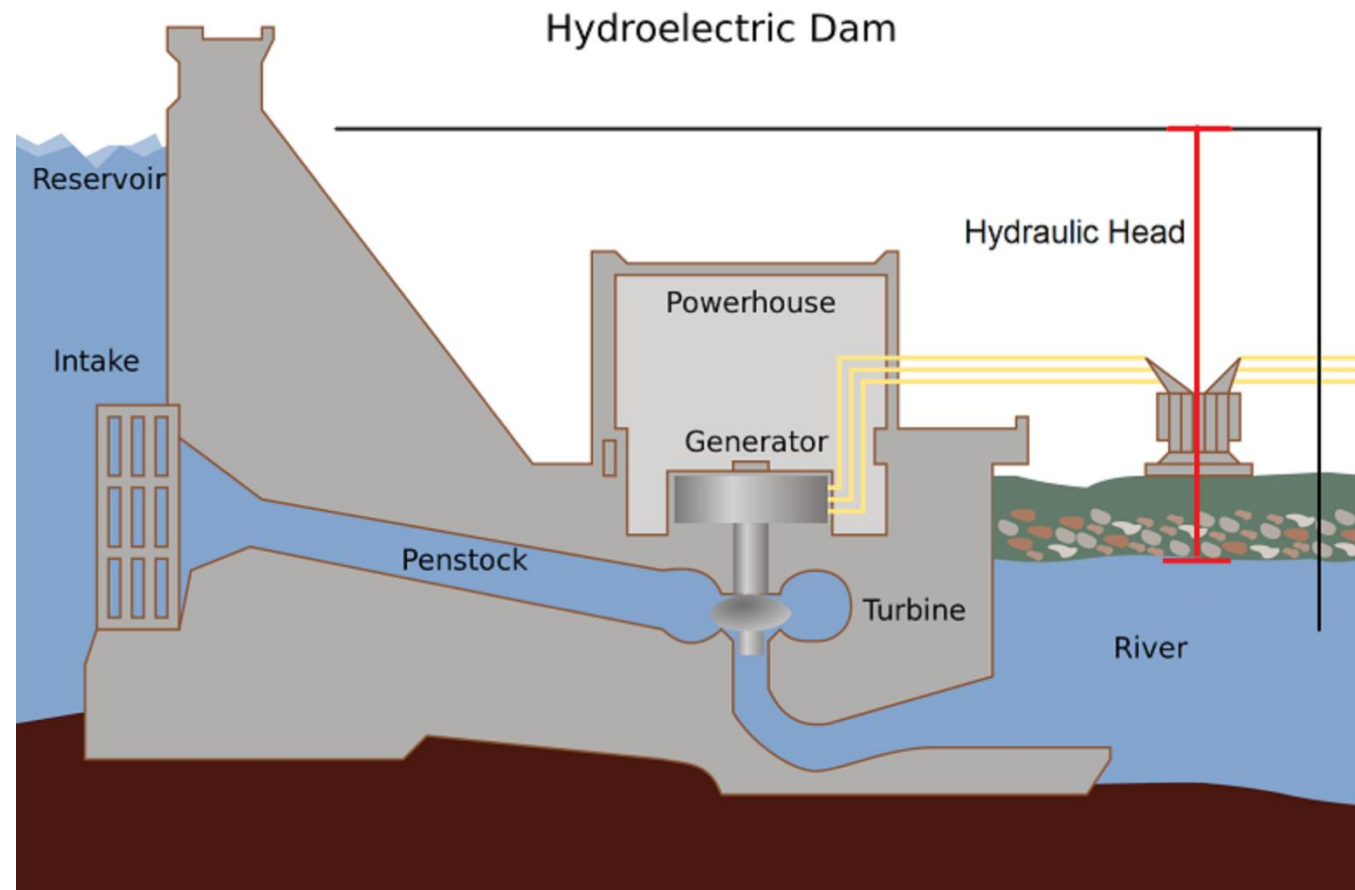
It should have the following:

- Background on the topic
- Some support from literature
- Clearly defined research question
- References
- Font size 11, justified paragraphs
- Figures, tables (properly cited with detailed captions)

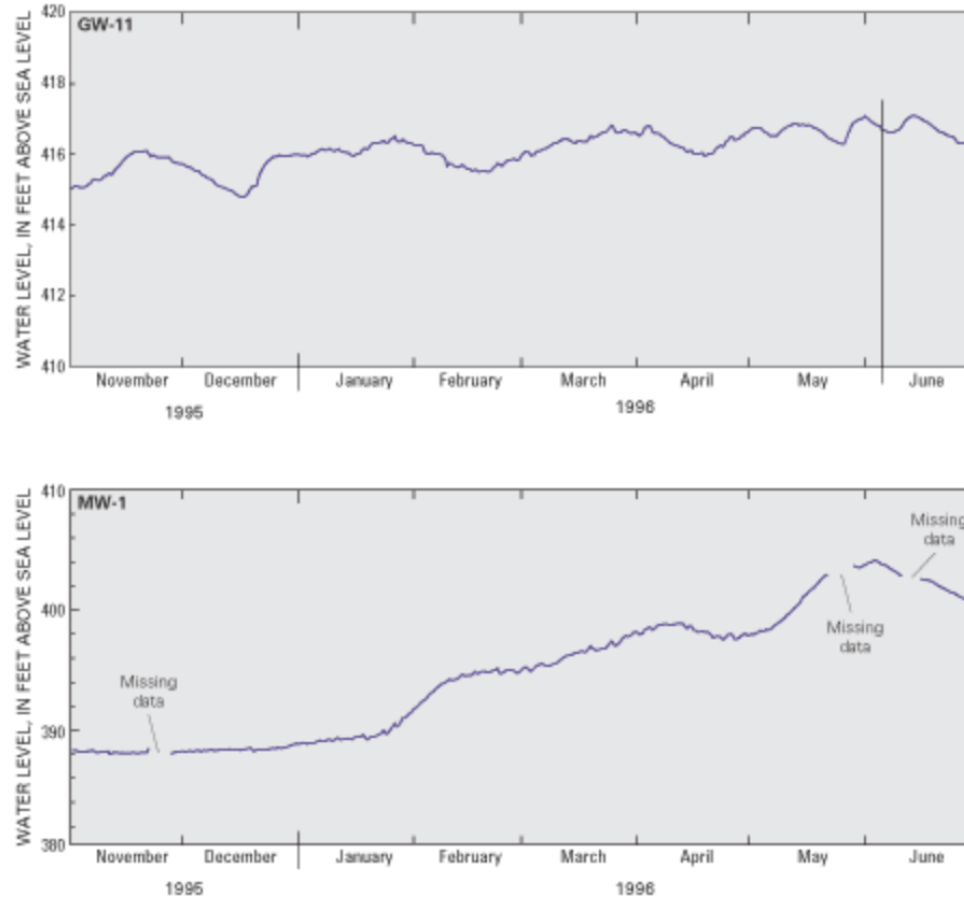
Hydraulic heads in wells



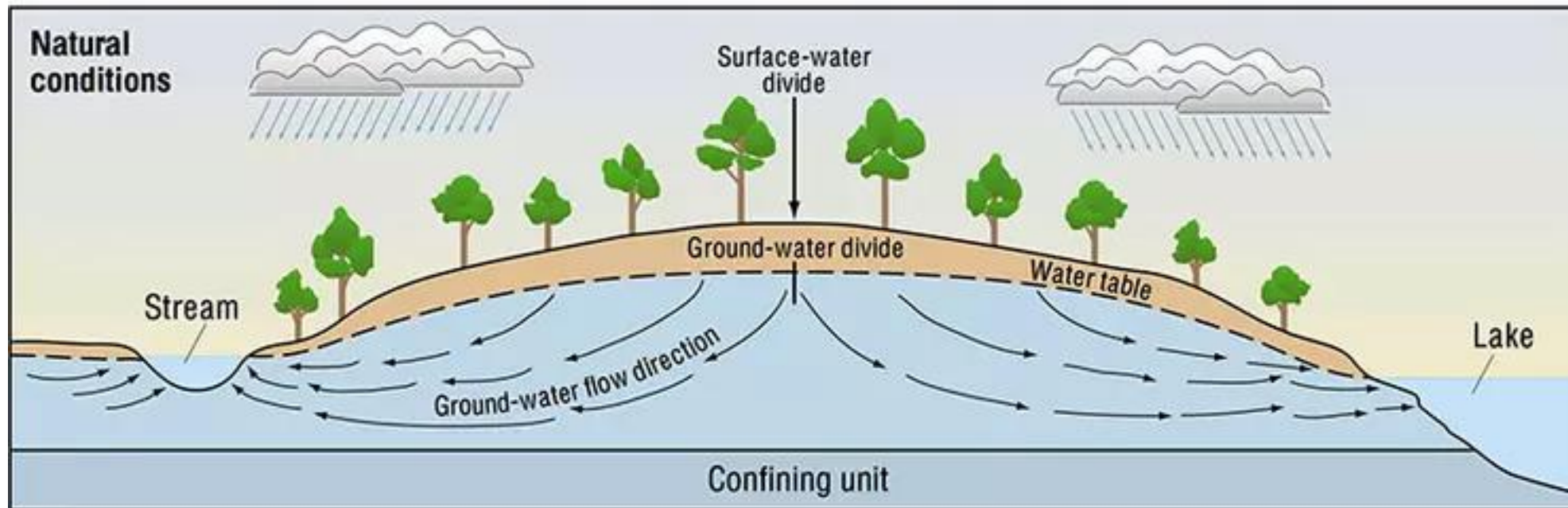
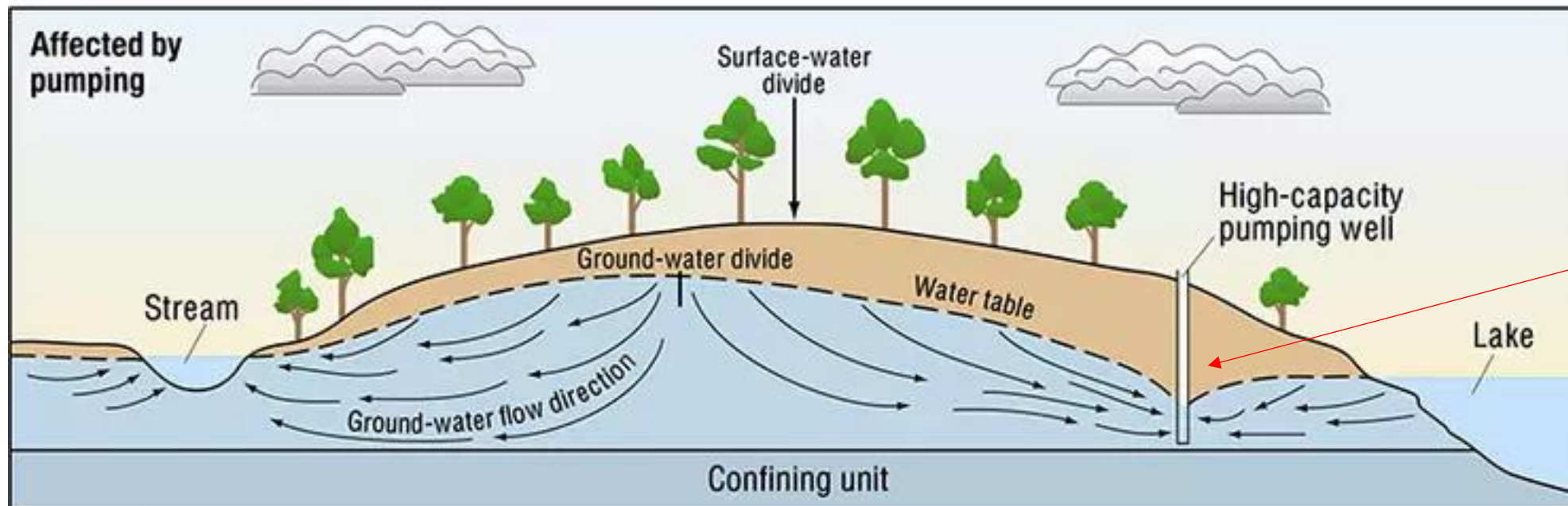
Hydraulic head



Seasonal variations

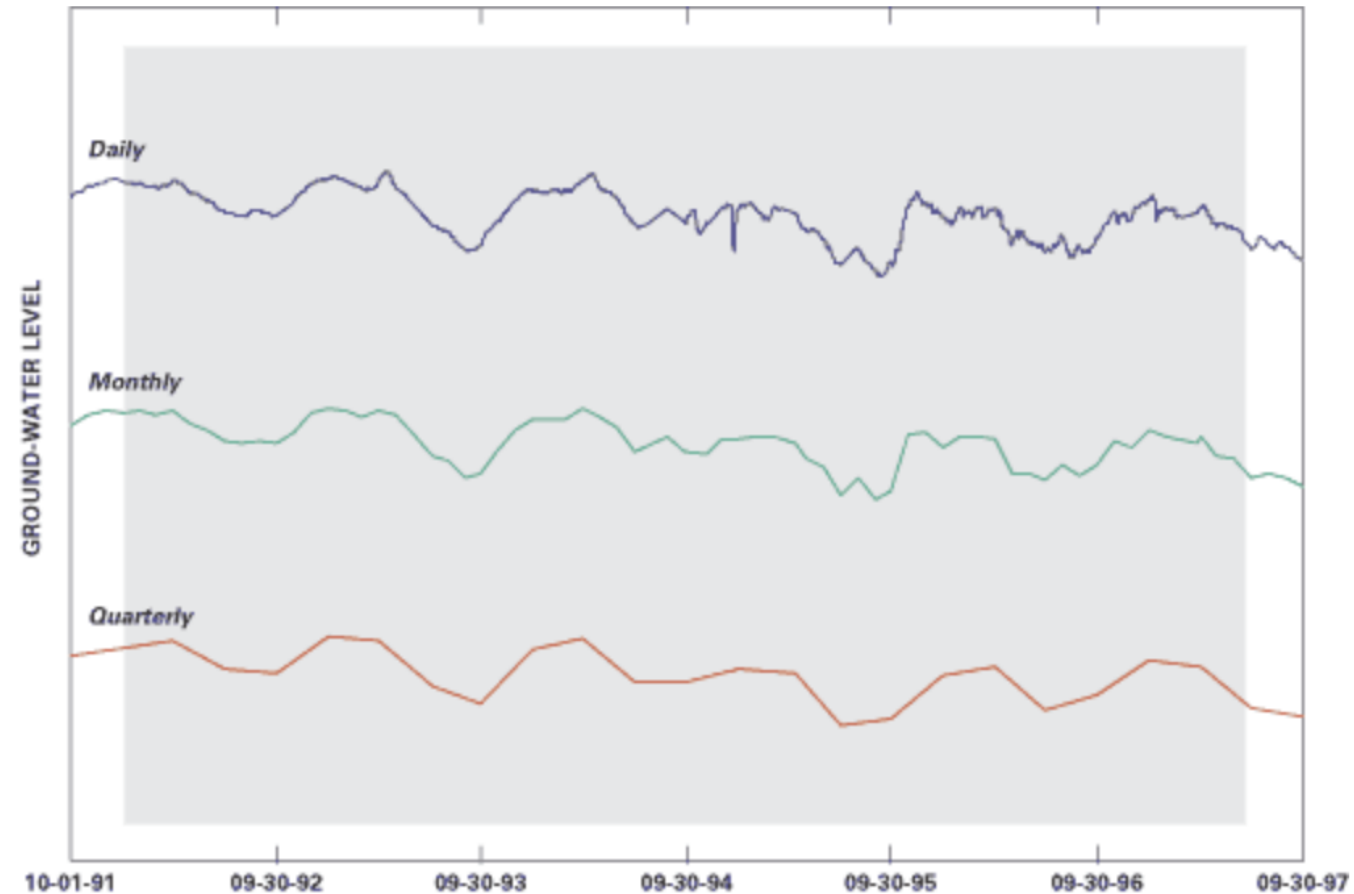


Example hydrographs showing the difference in timing and range of water-level fluctuations in two observation wells (GW-11 and MW-1)

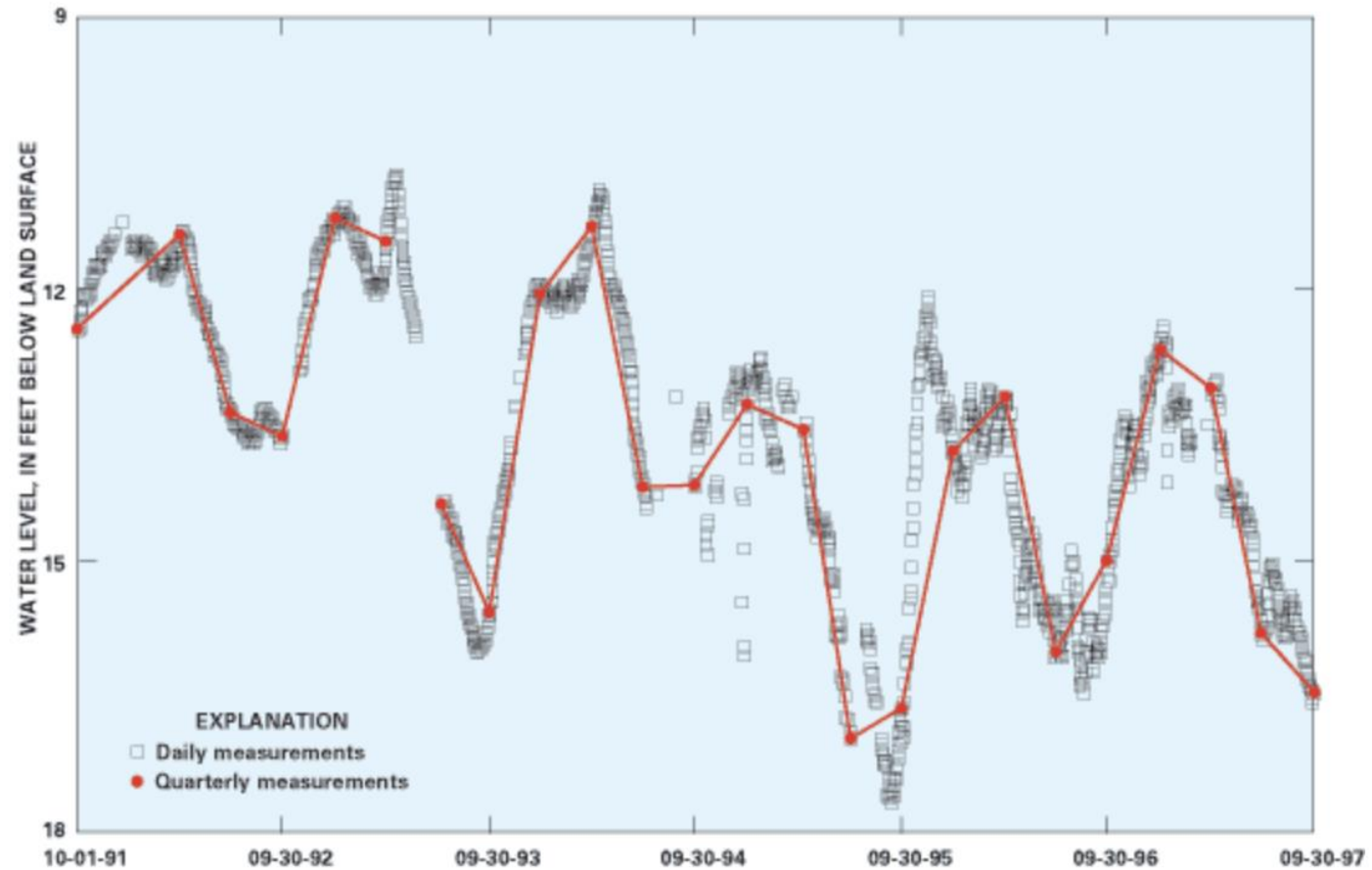
A**B**

Cone of depression

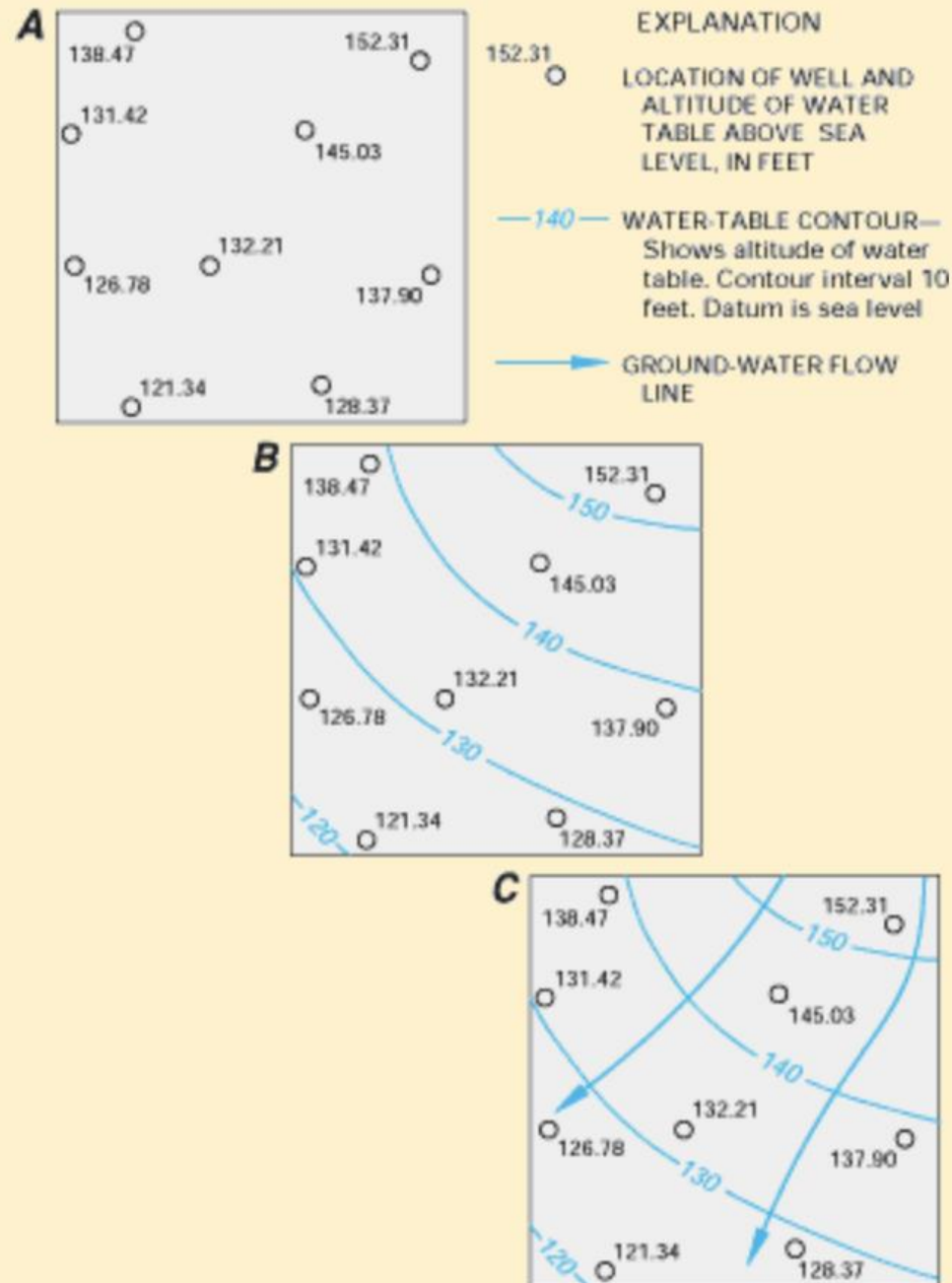
Water level measurement



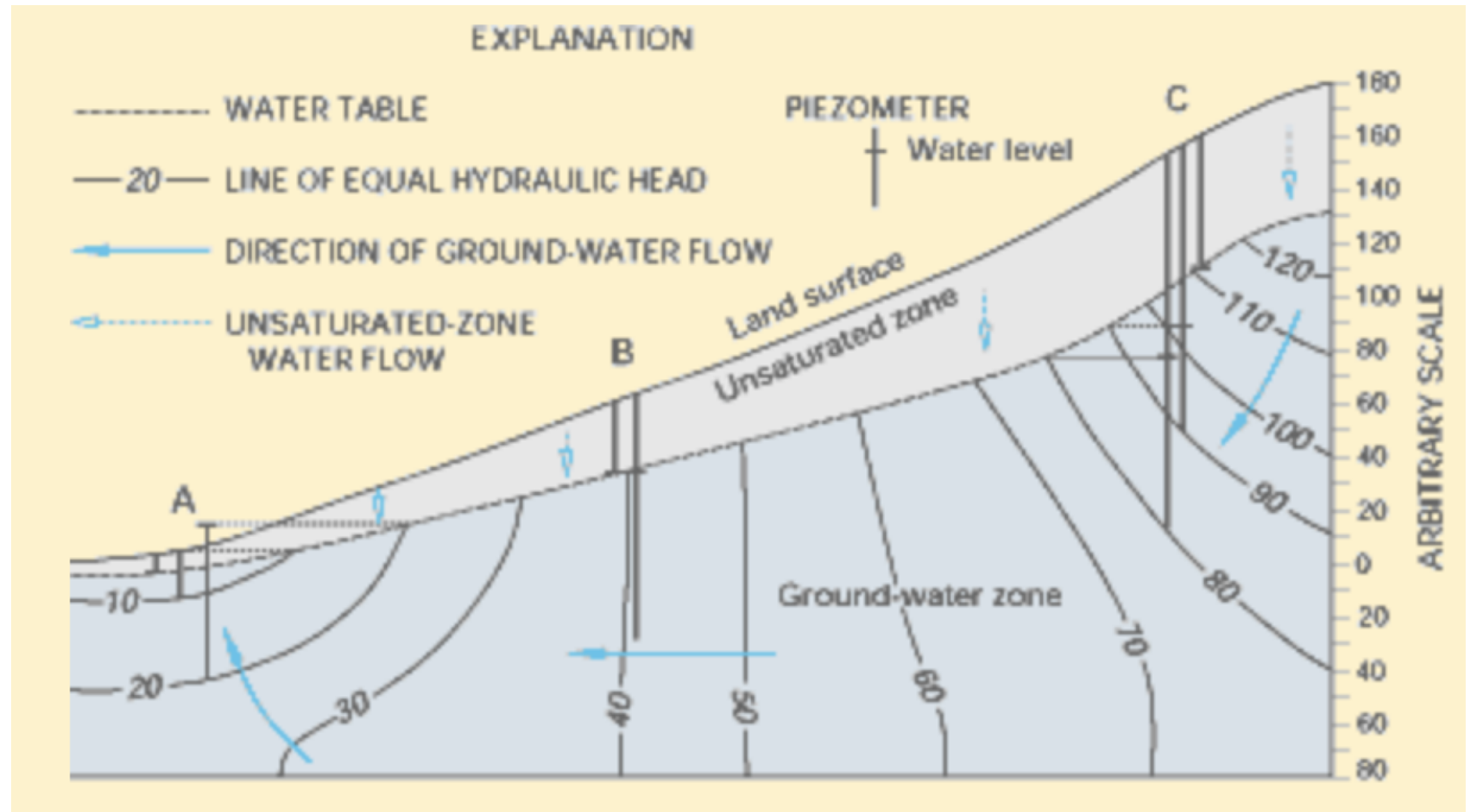
Water level measurement



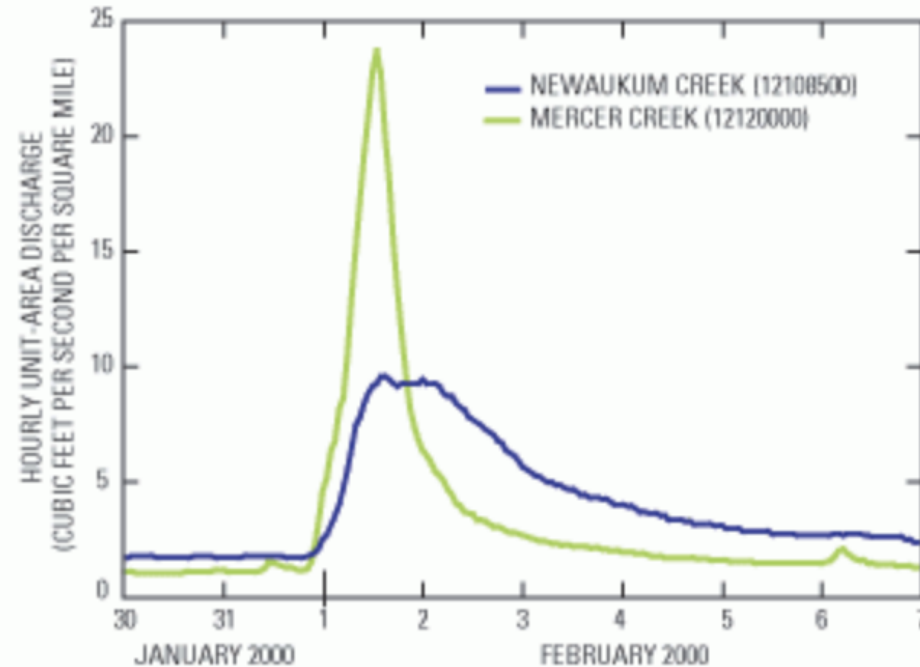
Water Table Contours



Water Table Contours



Surface Runoff & Urban Flooding



Streamflow in Mercer Creek, an urban stream in western Washington, increases more quickly, reaches a higher peak discharge, and has a larger volume during a one-day storm on February 1, 2000, than streamflow in Newaukum Creek, a nearby rural stream, a nearby rural stream that drains a basin of similar size.